

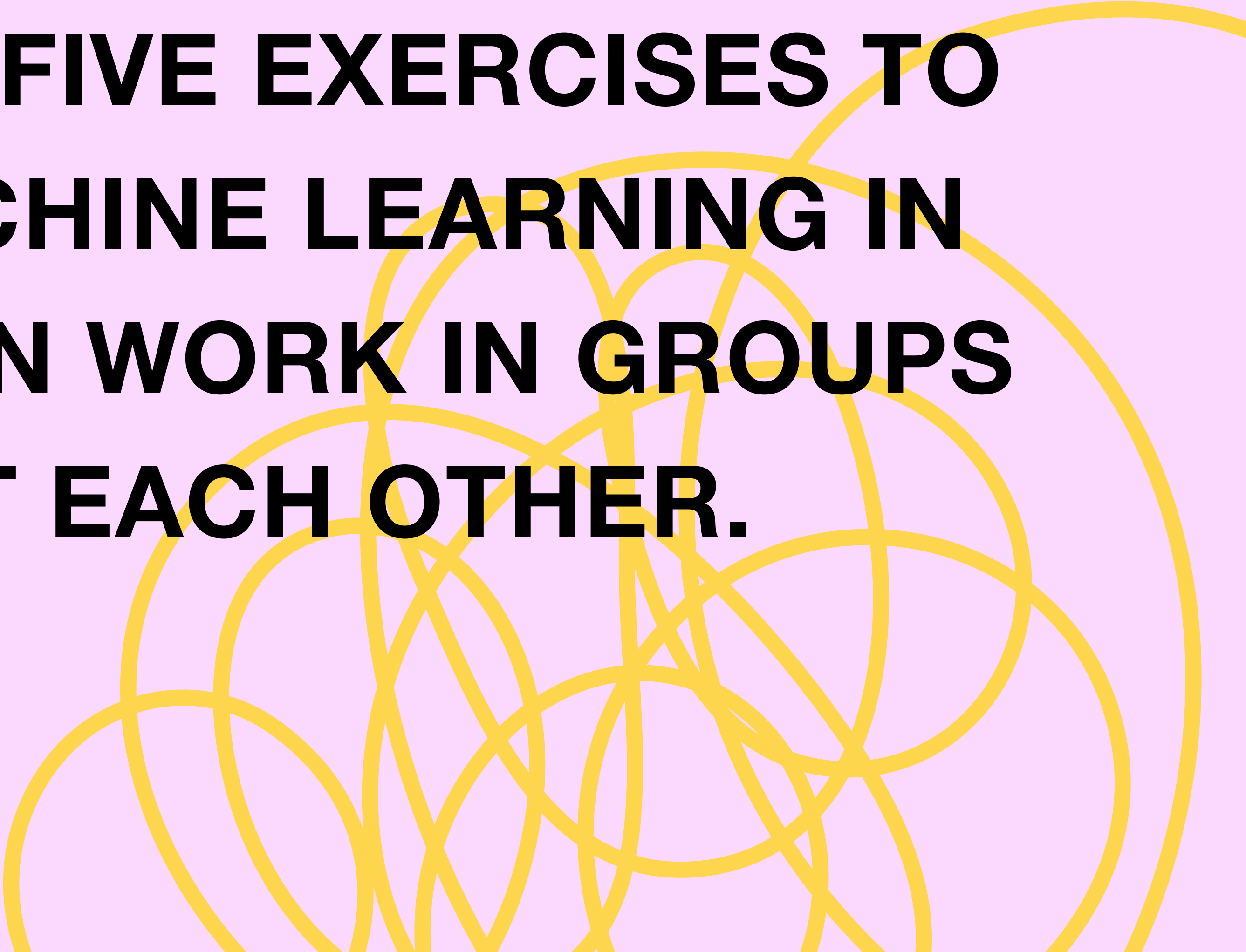


Complex Social &
Computational Systems

DIGGING INTO CODE – A ML EXERCISE IN PYTHON

A PSYCHOLOGIST'S GUIDE TO MACHINE LEARNING
ZURICH, MARCH 16-20, 2026

**CHOOSE ONE OF FIVE EXERCISES TO
IMPLEMENT MACHINE LEARNING IN
PYTHON. YOU CAN WORK IN GROUPS
TO SUPPORT EACH OTHER.**

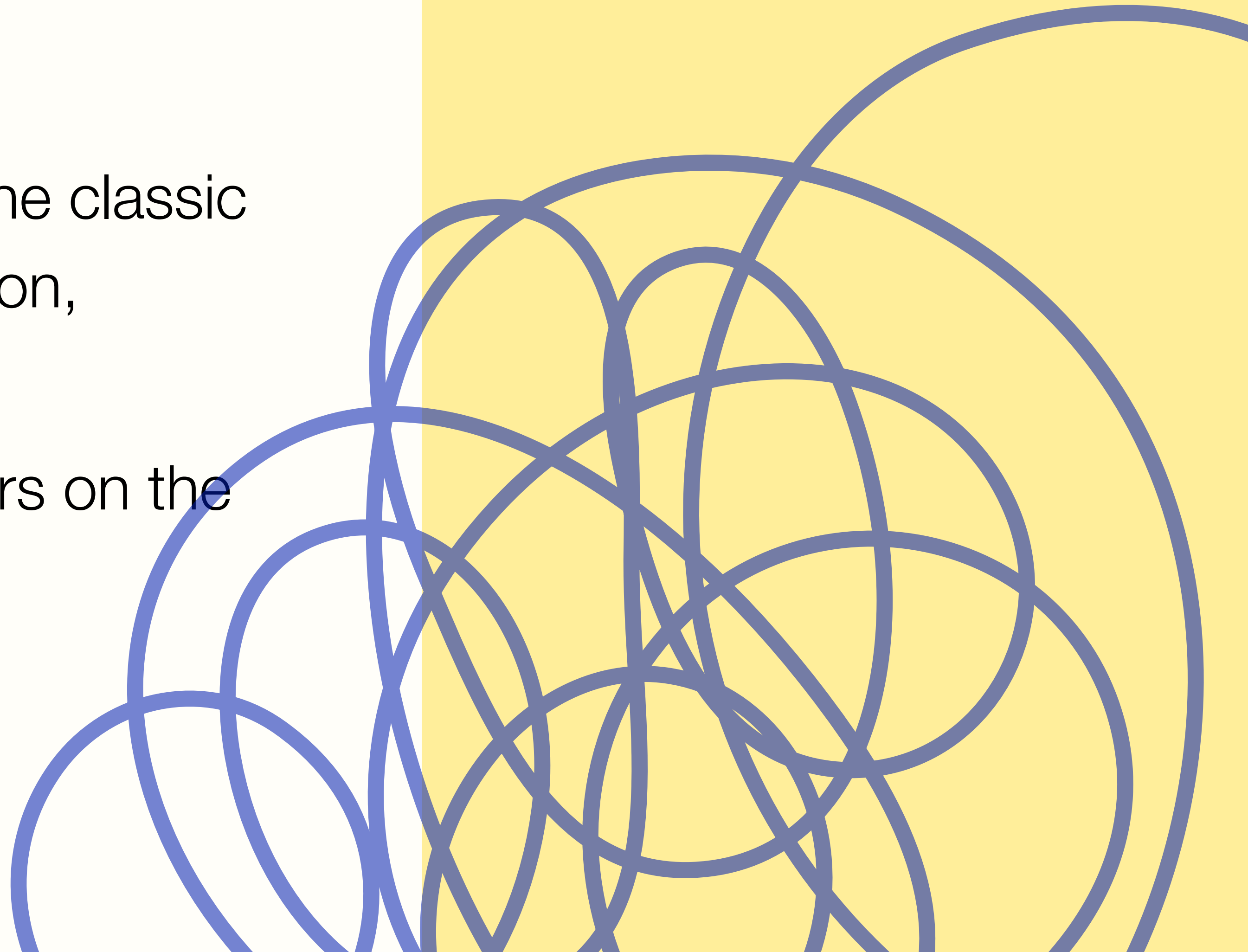
A series of overlapping yellow circles of varying sizes are positioned in the bottom right corner of the slide, partially overlapping the text area.

EX 1: SURVIVAL ON THE TITANIC

Expert level: ★ ★ ★

Keywords: machine learning basics, the classic data science pipeline, logistic regression, random forests

Goal: Predict the survival of passengers on the Titanic.

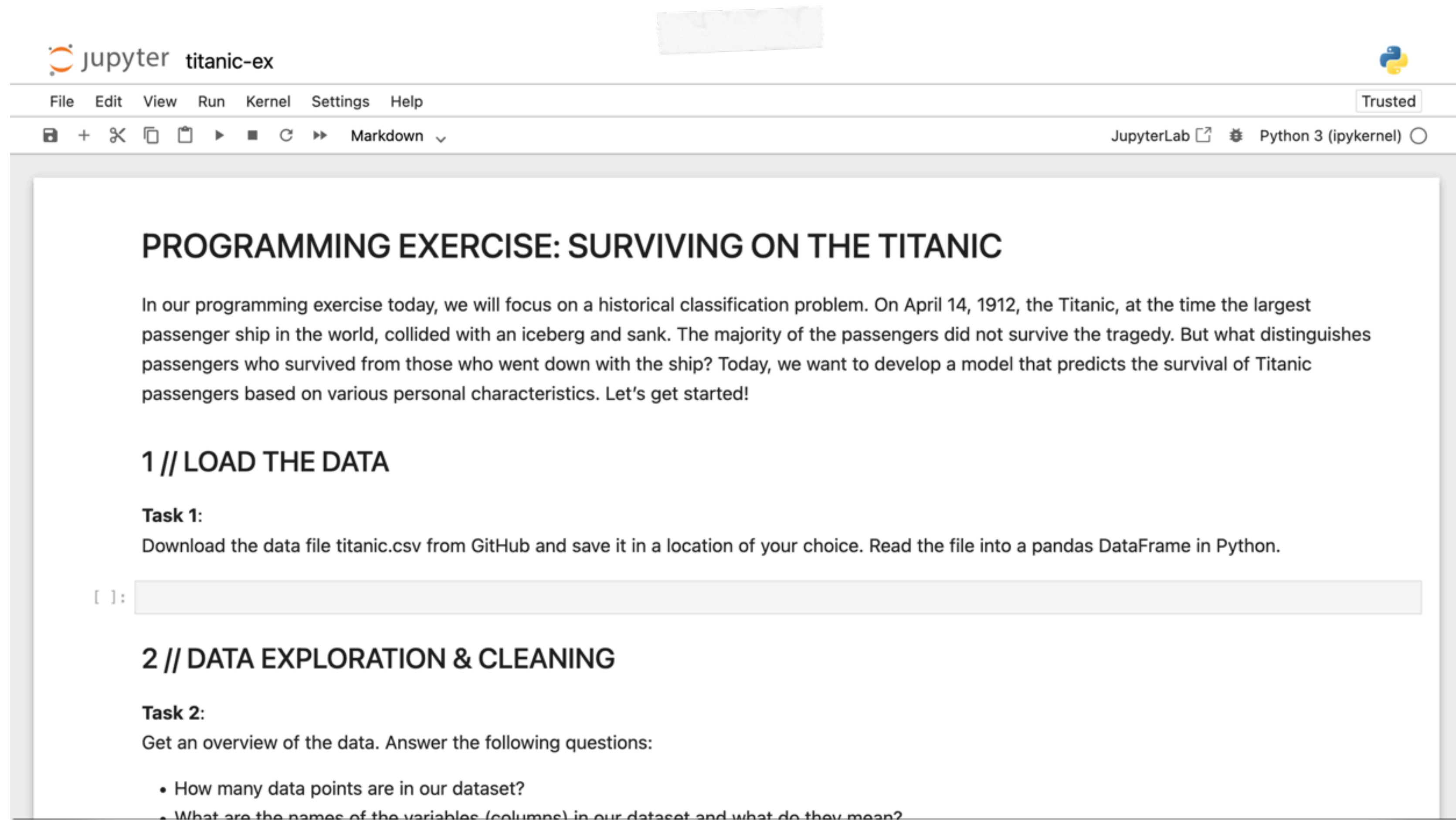


PREDICTING SURVIVAL ON THE TITANIC: THE DATA.

Data Dictionary

Variable	Definition	Key
survival	Survival	0 = No, 1 = Yes
pclass	Ticket class	1 = 1st, 2 = 2nd, 3 = 3rd
sex	Sex	
Age	Age in years	
sibsp	# of siblings / spouses aboard the Titanic	
parch	# of parents / children aboard the Titanic	
ticket	Ticket number	
fare	Passenger fare	
cabin	Cabin number	
embarked	Port of Embarkation	C = Cherbourg, Q = Queenstown, S = Southampton

PREDICTING SURVIVAL ON THE TITANIC: THE NOTEBOOK.



jupyter titanic-ex

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (ipykernel)

PROGRAMMING EXERCISE: SURVIVING ON THE TITANIC

In our programming exercise today, we will focus on a historical classification problem. On April 14, 1912, the Titanic, at the time the largest passenger ship in the world, collided with an iceberg and sank. The majority of the passengers did not survive the tragedy. But what distinguishes passengers who survived from those who went down with the ship? Today, we want to develop a model that predicts the survival of Titanic passengers based on various personal characteristics. Let's get started!

1 // LOAD THE DATA

Task 1:
Download the data file `titanic.csv` from GitHub and save it in a location of your choice. Read the file into a pandas DataFrame in Python.

```
[ ]:
```

2 // DATA EXPLORATION & CLEANING

Task 2:
Get an overview of the data. Answer the following questions:

- How many data points are in our dataset?
- What are the names of the variables (columns) in our dataset and what do they mean?

EX 2: EXPLORING MENTAL HEALTH DATA

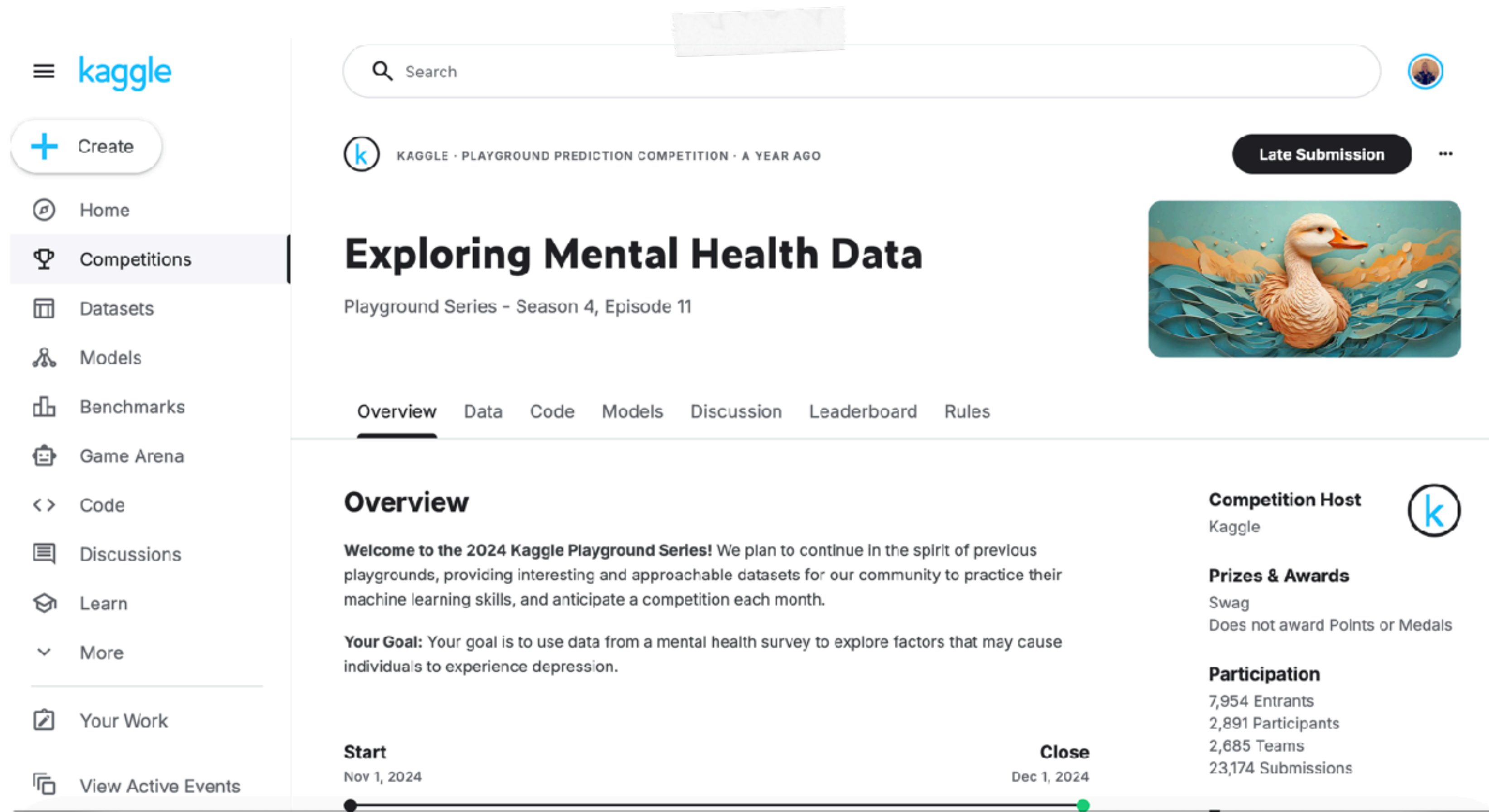
Expert level: ★ ★ ★

Keywords: Kaggle playground, synthetic data, classification, data visualization, feature engineering, ensemble learning

Goal: Use data from mental health survey to predict depression.



EXPLORING MENTAL HEALTH DATA: A PLAYGROUND COMPETITION ON KAGGLE.



The screenshot shows the Kaggle Playground Prediction Competition page for 'Exploring Mental Health Data'. The page is titled 'Exploring Mental Health Data' and is part of the 'Playground Series - Season 4, Episode 11'. The competition is hosted by Kaggle and is currently in progress, with a 'Late Submission' button visible. The page includes a sidebar with navigation links, a search bar, and a main content area with tabs for Overview, Data, Code, Models, Discussion, Leaderboard, and Rules. The Overview tab is selected, showing a welcome message and the competition's goal. A timeline at the bottom indicates the start date (Nov 1, 2024) and the close date (Dec 1, 2024).

Kaggle

Search

KAGGLE · PLAYGROUND PREDICTION COMPETITION · A YEAR AGO

Exploring Mental Health Data

Playground Series - Season 4, Episode 11

Late Submission

Overview Data Code Models Discussion Leaderboard Rules

Overview

Welcome to the 2024 Kaggle Playground Series! We plan to continue in the spirit of previous playgrounds, providing interesting and approachable datasets for our community to practice their machine learning skills, and anticipate a competition each month.

Your Goal: Your goal is to use data from a mental health survey to explore factors that may cause individuals to experience depression.

Competition Host
Kaggle

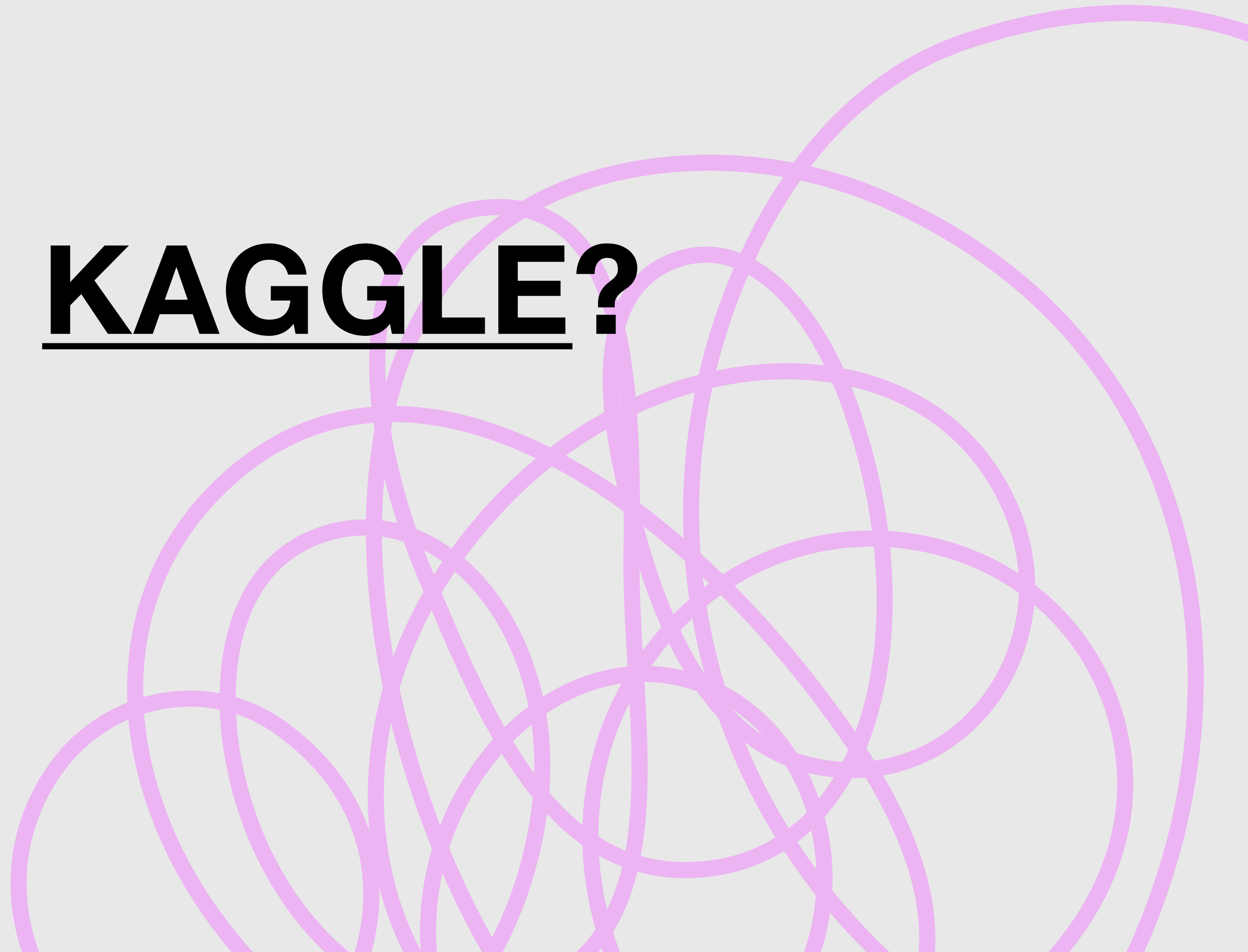
Prizes & Awards
Swag
Does not award Points or Medals

Participation
7,954 Entrants
2,891 Participants
2,685 Teams
23,174 Submissions

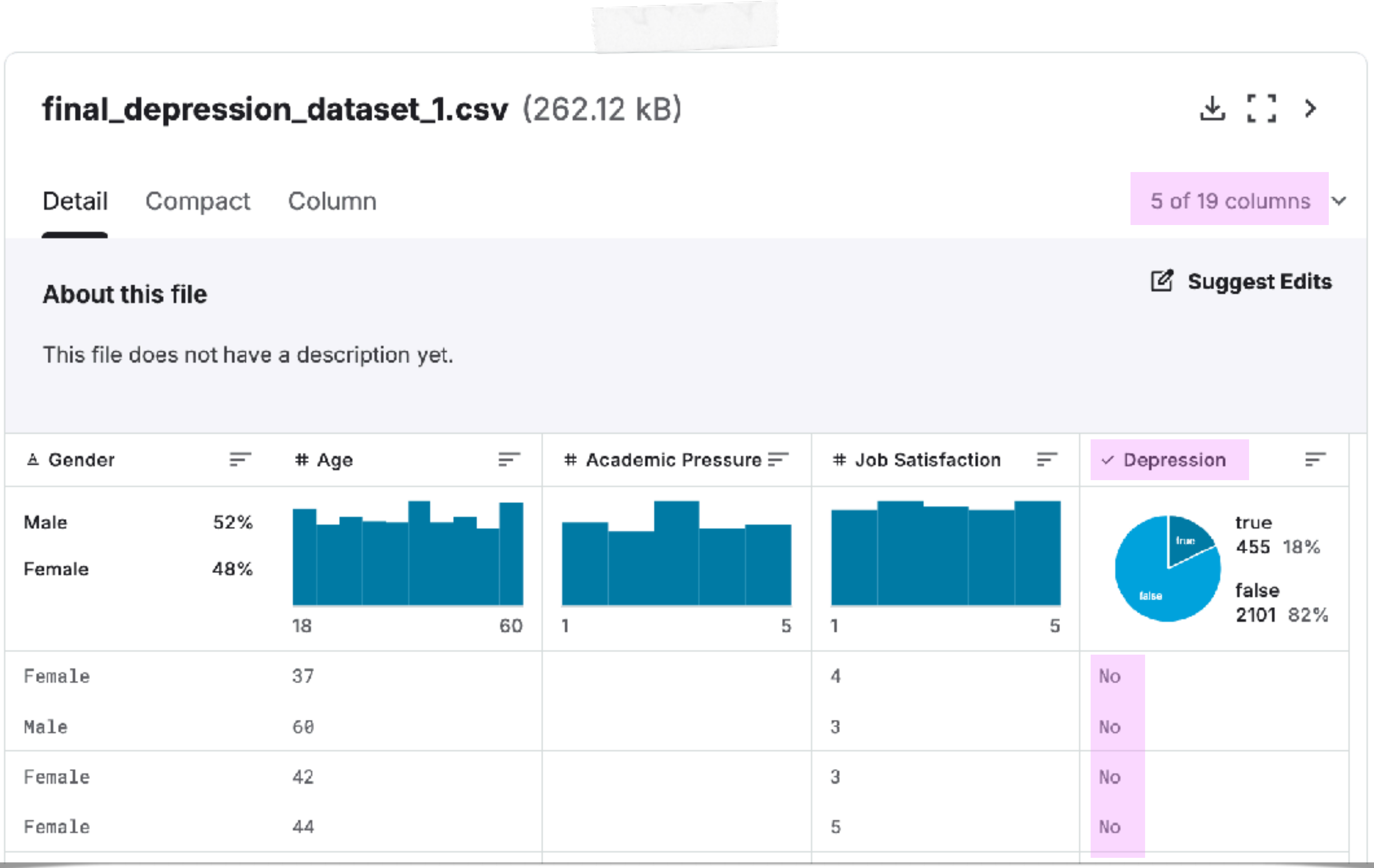
Start
Nov 1, 2024

Close
Dec 1, 2024

WHAT IS KAGGLE?



EXPLORING MENTAL HEALTH DATA: THE DATA.

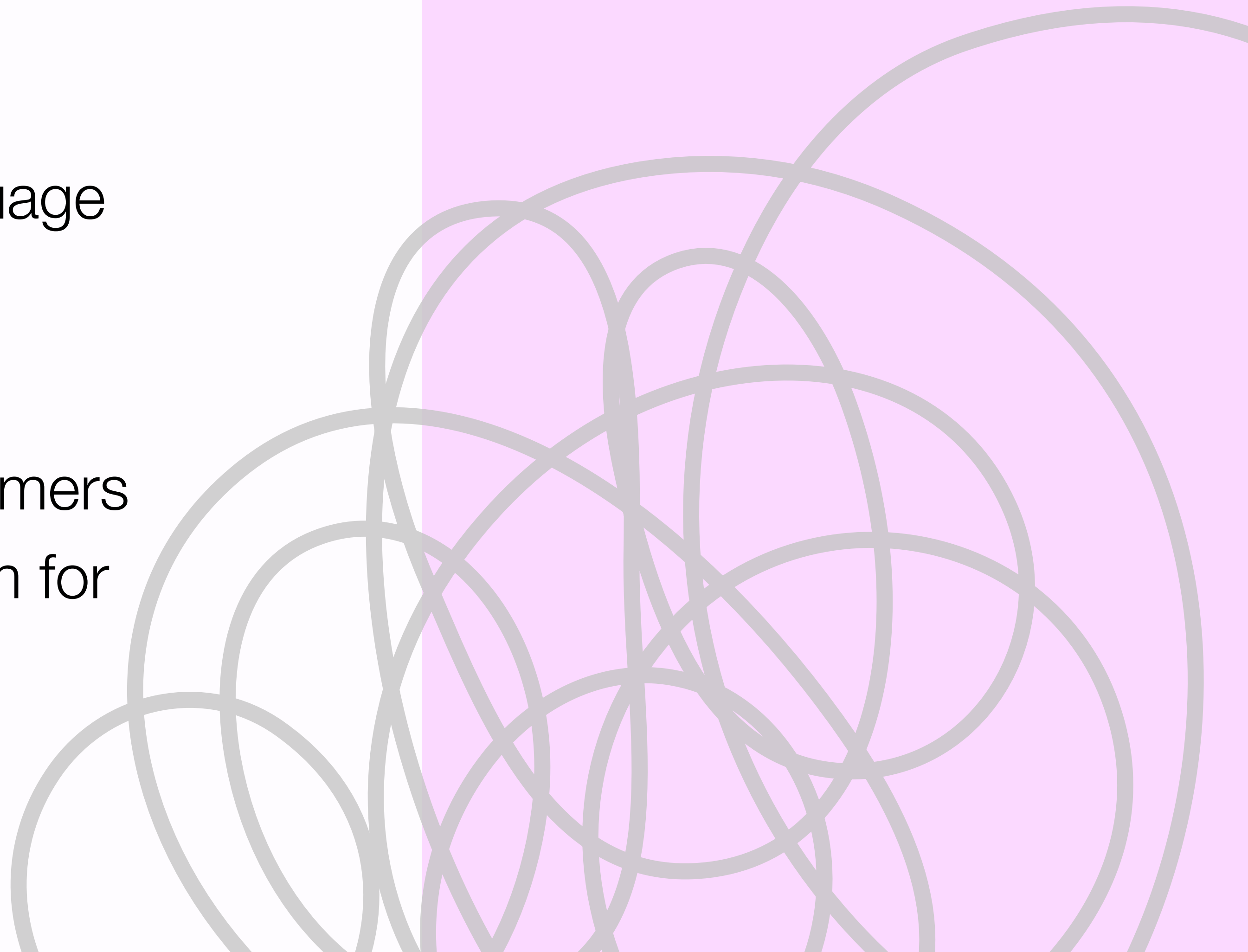


EX 3: THE HUGGING FACE LLM COURSE

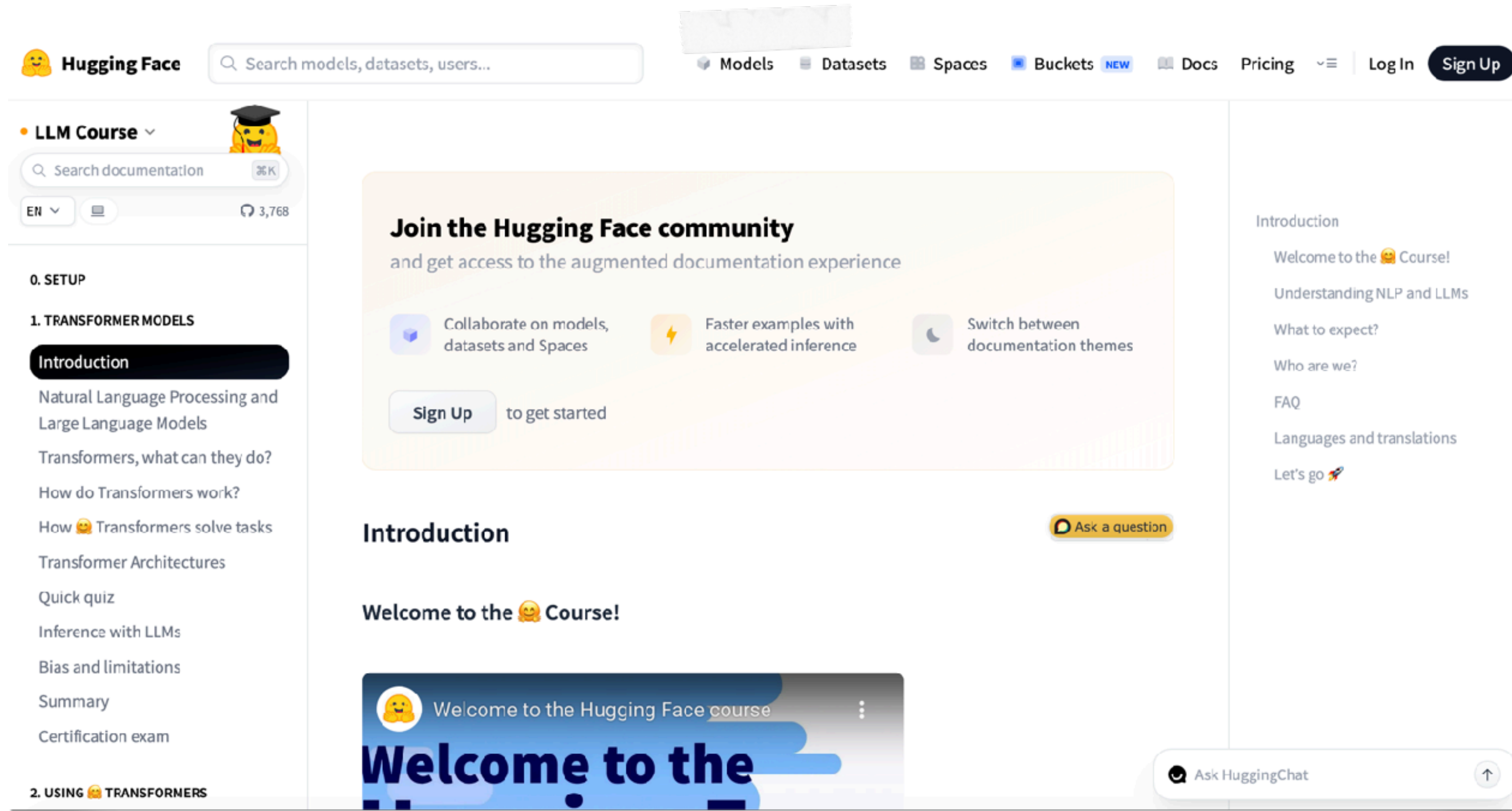
Expert level: ★ ★ ★

Keywords: Huggingface, natural language processing, transformers, LLMs, text classification

Goal: Familiarize yourself with transformers (LLMs), how to use and fine-tune them for different tasks.



THE HUGGINGFACE LLM COURSE: WHAT IS HUGGINGFACE?



The screenshot shows the Hugging Face website's LLM Course page. The top navigation bar includes the Hugging Face logo, a search bar, and links to Models, Datasets, Spaces, Buckets, Docs, Pricing, Log In, and Sign Up. The left sidebar features the 'LLM Course' dropdown menu with a search bar, language selection (EN), and a list of course topics. The main content area has a large yellow banner for joining the community, followed by an 'Introduction' section with a welcome message and a video player. The right sidebar lists the course topics. The bottom right corner has a chat button labeled 'Ask HuggingChat'.

Hugging Face Search models, datasets, users...

Models Datasets Spaces Buckets **NEW** Docs Pricing Log In Sign Up

LLM Course Search documentation 3,768

EN

0. SETUP

1. TRANSFORMER MODELS

Introduction

Natural Language Processing and Large Language Models

Transformers, what can they do?

How do Transformers work?

How Transformers solve tasks

Transformer Architectures

Quick quiz

Inference with LLMs

Bias and limitations

Summary

Certification exam

2. USING TRANSFORMERS

Join the Hugging Face community
and get access to the augmented documentation experience

Collaborate on models, datasets and Spaces

Faster examples with accelerated inference

Switch between documentation themes

Sign Up to get started

Introduction Ask a question

Welcome to the Course!

Welcome to the Hugging Face course

Welcome to the

Ask HuggingChat

Introduction

Welcome to the Course!

Understanding NLP and LLMs

What to expect?

Who are we?

FAQ

Languages and translations

Let's go

THE HUGGINGFACE LLM COURSE: THE CONTENT.

What to expect?

Here is a brief overview of the course:

Introduction

Transformer models

Using 🤗 Transformers

Fine-tuning a pretrained model

Sharing models and tokenizers

Diving in

The 🤗 Datasets library

The 🤗 Tokenizers library

Main NLP tasks

How to ask for help

Advanced

Building and sharing demos

Transformers can hear

Transformers can see

Optimizing for production

start here

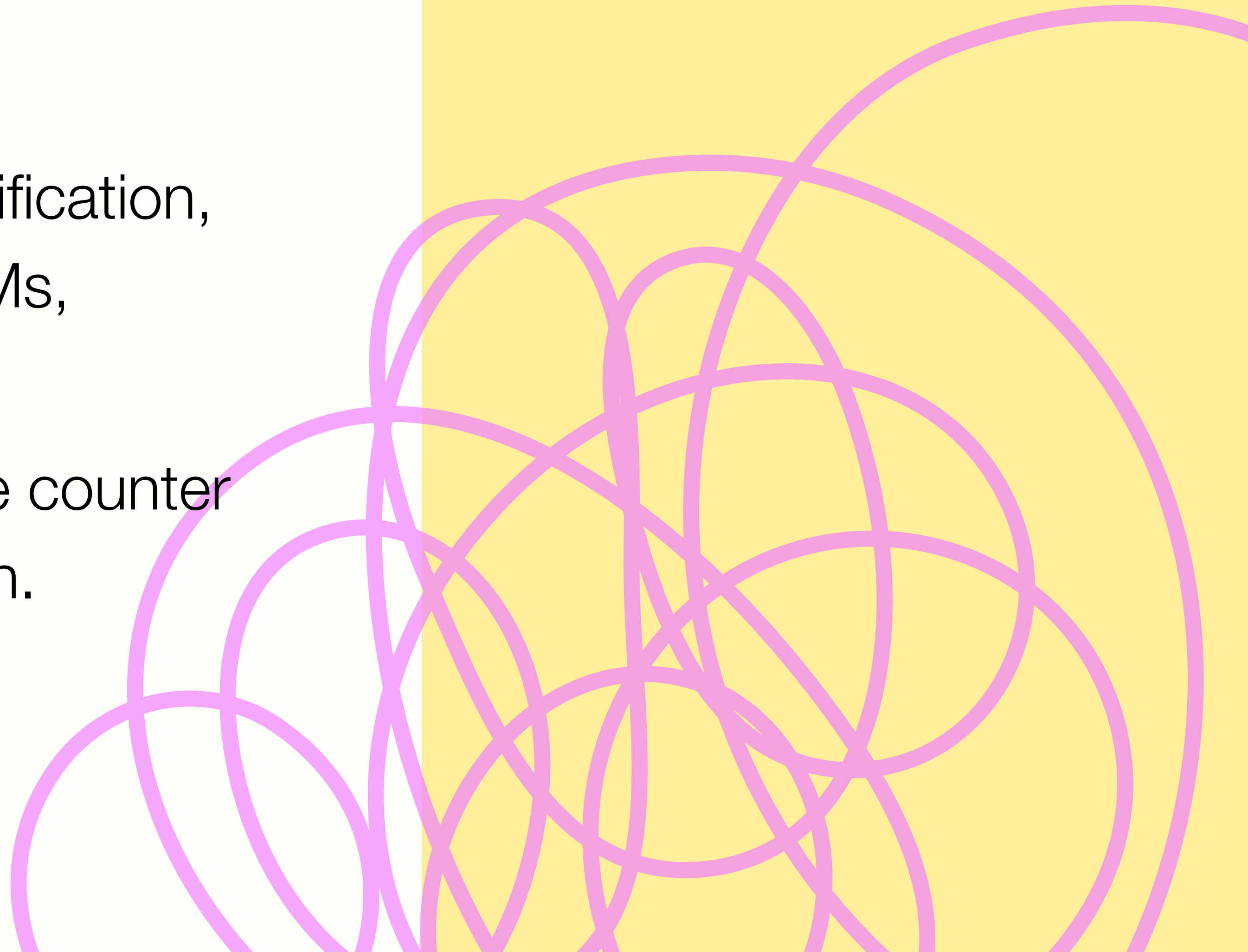
probably not for today

EX 4: COMPUTATIONAL SOCIAL MIXED METHODS

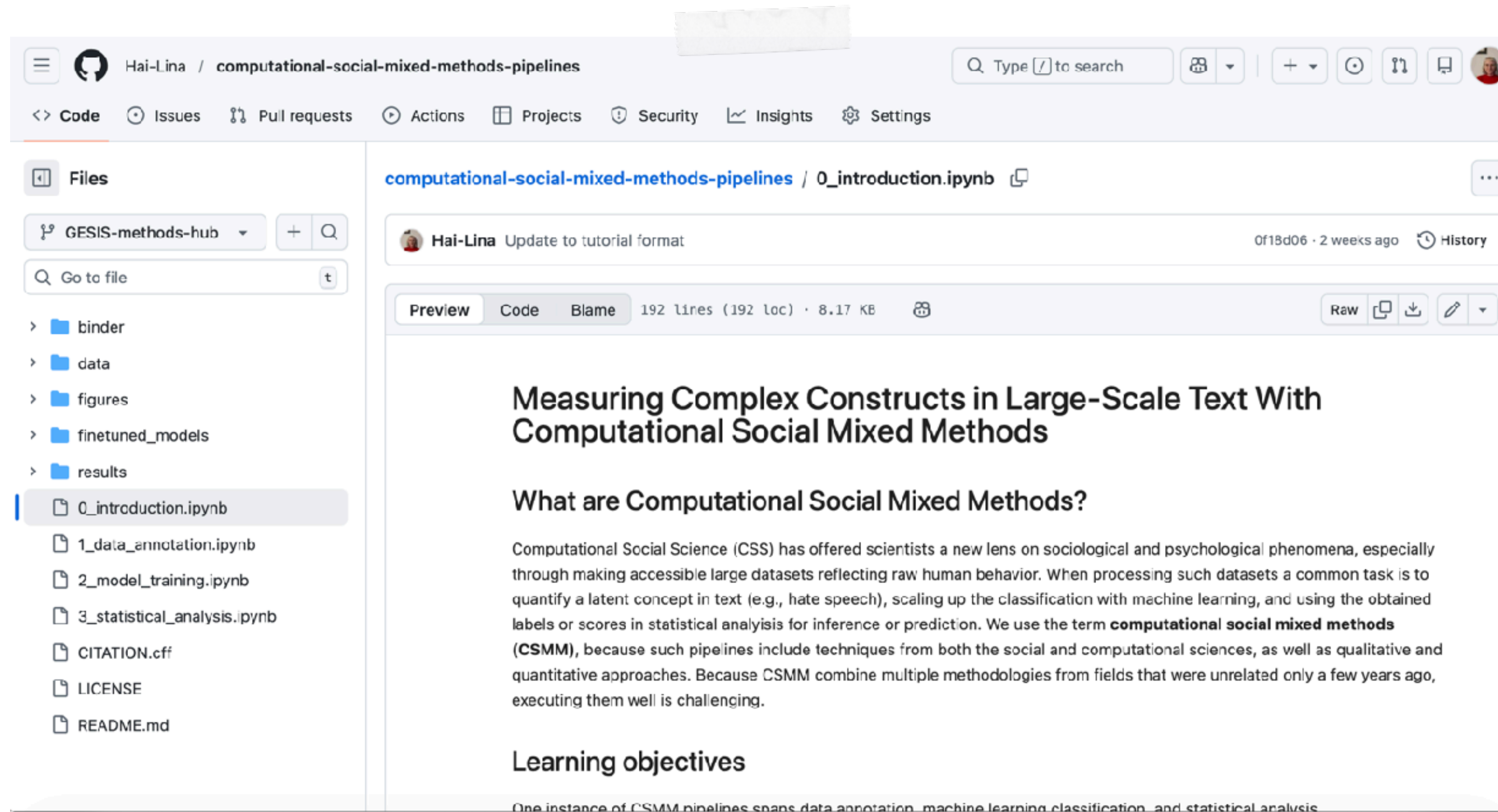
Expert level: ★ ★ ★

Keywords: data annotation, text classification,
statistical modeling, transformers, LLMs,
advanced training techniques

Goal: Reproduce our pipeline from the counter
speech paper in a tutorial-style fashion.



COMPUTATIONAL SOCIAL MIXED METHODS: A TUTORIAL



COMPUTATIONAL SOCIAL MIXED METHODS: WHAT ARE CSMM?

CONSTRUCTS IN TEXT GUIDELINES1

Measuring complex constructs in large-scale text with computational social mixed methods

Alina Herderich^{1,2}, Jona Lasser^{1,3}, Mirta Galešić^{3,4,5}, Segun Aroyebun⁶, David Garcia^{2,3,6}, and Joshua Garland⁷

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²Department of Computer Science and Biomedical Engineering, Graz University of Technology

³Complexity Science Hub Vienna

⁴Santa Fe Institute

⁵Vermont Complex Systems Center, University of Vermont

⁶Department of Politics and Public Administration, University of Konstanz

⁷Center on Information and Narrative Complexity, Arizona State University

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Link to preprint: <https://psyarxiv.com/tzc9p> (doi: 10.31234/osf.io/tzc9p)

1. DATA ANNOTATION

Developing a valid classification scheme

Composing the annotation team

Sampling the ideal training data

 *What if... the training data is unbalanced?*
ITERATIVE ANNOTATION

2. MODEL TRAINING

Selecting an adequate base model

Evaluating model performance

Improving model performance

 *What if... model performance plateaus?*
TRAINING ON CONFIDENT EXAMPLES

3. STATISTICAL ANALYSIS

Accounting for classifier inaccuracy

 *What if... statistical analysis involves group comparisons?*
EMPIRICAL THRESHOLDS FOR BINARIZING CLASSIFIER BASED BOOTSTRAPPING

EX 5: CHOOSE YOUR OWN ADVENTURE

Expert level: ★? ★? ★?

Keywords: bring your own data

Goal: Get started on your personal machine learning project with your own data. You can consult me on how and where to start.

Come talk to me one on one!





HOW TO CREATE A VENV FROM FILE


```
programming-ex — -zsh — 80x24

Last login: Wed Mar 11 16:56:54 on ttys000
[(base) alinaherderich@Mac ~ % cd /Users/alinaherderich/SynologyDrive/Post-Doc/te]
aching/extern/workshop-materials/programming-ex
[(base) alinaherderich@Mac programming-ex % ls ]
psych-ML.yml          titanic-ex.ipynb
titanic-ex-solution.ipynb  titanic.csv
(base) alinaherderich@Mac programming-ex % conda env create -f psych-ML.yml
```

09:30 - 12:30

CODING PRACTICALS I

LUNCH BREAK AFTERWARDS